Solutions to Select Problems From *Hall. Lie Groups, Lie Algebras, and Representations: An Elementary Introduction.*

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Preface

This document is a compilation of exercises I have completed from the second edition of Brian C. Hall's *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction.* There are several motivations behind this project. For one, I was unable to take a course on representations of groups offered at my undergraduate institution. Moreover, Lie Groups and Algebras occur often within physics, and I am interested in studying their mathematical foundation.

The textbook was chosen based on a combination of its rigor, and reputation as particularly helpful for students of physics. The exercises shown here were not chosen in any systematic fashion.

I have used the Einstein summation convention throughout the document unless otherwise specified. As of this writing, the project remains in progress.

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1 Matrix Lie Groups

1. Let a be an irrational real number and let G be the following subgroup of $\mathsf{GL}(2,\mathbb{C})$:

$$G = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{ita} \end{pmatrix} \middle| \, t \in \mathbb{R} \right\}.$$

Show that

$$\overline{G} = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{is} \end{pmatrix} \middle| \, t, s \in \mathbb{R} \right\},$$

where $\overline{G}$ denotes the closure of the set G inside the space of 2×2 matrices. Assume that the set of numbers of the form $e^{2\pi i n a}, n \in \mathbb{Z}$, is dense in S^1.

Proof. Take some $M \in \overline{G}$. By the definition of the closure, there exists a sequence $M_n \to M$, with $M_n \in G$. We can explicitly write M_n as

$$M_n = \begin{pmatrix} e^{it_n} & 0 \\ 0 & e^{it_n a} \end{pmatrix}$$

for some sequence of real numbers t_n. The set S^1 is clearly closed within $\mathbb{C}$, so that $e^{it_n} \to e^{it}$ for some $t \in \mathbb{R}$ (defined up to a multiple of 2π) and analogously $e^{it_n a} \to e^{is}$ for some $s \in \mathbb{R}$. We clearly have the off-diagonal elements as tending to 0 since they are always 0, and hence we have

$$M = \begin{pmatrix} e^{it} & 0 \\ 0 & e^{is} \end{pmatrix},$$

as desired. Conversely, take some $t_0, s_0 \in \mathbb{R}$ to form the matrix

$$M = \begin{pmatrix} e^{it_0} & 0 \\ 0 & e^{is_0} \end{pmatrix}.$$

By the described density property, there exists a sequence $t_n \in \mathbb{Z}$ such that $e^{2\pi i t_n a} \to e^{i(s_0 - t_0 a)}$. Then consider the sequence described by

$$M_n = \begin{pmatrix} e^{i(t_0 + 2\pi i t_n)} & 0 \\ 0 & e^{i(t_0 + 2\pi i t_n)a} \end{pmatrix} = \begin{pmatrix} e^{it_0} & 0 \\ 0 & e^{it_0 a} e^{2\pi i t_n a} \end{pmatrix}.$$

We have $M_n \in G$ with $t := t_0 + 2\pi i t_n \in \mathbb{R}$. With e^{it_0} and $e^{it_0 a}$ as constants, we have

$$M_n \to \begin{pmatrix} e^{it_0} & 0 \\ 0 & e^{it_0 a} e^{i(s_0 - t_0 a)} \end{pmatrix} = \begin{pmatrix} e^{it_0} & 0 \\ 0 & e^{is_0} \end{pmatrix} = M,$$

so that $M \in \overline{G}$ and the desired equivalence is shown. ∎

2. *Orthogonal groups.* Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^n : \langle x, y \rangle = x_k y_k$. Show that a matrix A preserves the inner product if and only if the column vectors of A are orthonormal.
Show that for any $n \times n$ real matrix B,

$$\langle Bx, y \rangle = \langle x, B^{\mathrm{T}} y \rangle$$

where $(B^{\mathrm{T}})_{kl} = B_{lk}$. Using this, show that a matrix A preserves the inner product on $\mathbb{R}^n$ if and only if $A^{\mathrm{T}} A = I$.

Proof. Suppose a matrix A preserves the inner product for all $x, y \in \mathbb{R}^n$. We can write this as

$$x_i y_i = \langle x, y \rangle = \langle Ax, Ay \rangle = a_{lj} a_{lk} x_j y_k$$

Taking $x = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ yields $y_1 = a_{l1} a_{lk} y_k$ which is satisfied for arbitrary y if and only if $a_{l1} a_{lk} = \delta_{1k}$. The argument can be repeated with the position of the 1 in x transferred, such that $a_{lj} a_{lk} = \delta_{jk}$ and hence A has orthonormal column vectors.

Conversely, suppose the column vectors of A are orthonormal. Then

$$\langle Ax, Ay \rangle = a_{lj} a_{lk} x_j y_k = \delta_{jk} x_j y_k = x_i y_i = \langle x, y \rangle$$

as desired. Now take some $n \times n$ real matrix B. We have

$$\langle Bx, y \rangle = b_{kl} x_l y_k = b_{lk} x_k y_l = \langle x, B^{\mathrm{T}} y \rangle$$

since the variable used in the summation index is arbitrary and only the relative position matters. Thus, if a matrix A preserves the inner product,

$$\langle x, y \rangle = \langle Ax, Ay \rangle = \langle x, A^{\mathrm{T}} A y \rangle.$$

We can see this is satisfied only if $A^{\mathrm{T}} A y = y$ or equivalently $A^{\mathrm{T}} A = I$, since if there were an index in which y and $A^{\mathrm{T}} A y$ differ, we can construct a vector x which vanishes everywhere except that index and derive a contradiction. ■

3. *Unitary groups.* Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{C}^n : \langle x, y, \rangle = \overline{x_k} y_k$. Following Exercise 2, show that $\langle Ax, Ay \rangle = \langle x, y \rangle$ for all $x, y \in \mathbb{C}^n$ if and only if $A^* A = I$ and that this holds if and only if the columns of A are orthonormal.

Proof. Suppose a matrix A preserves the inner product for all $x, y \in \mathbb{C}^n$. We can write this as

$$\overline{x_i} y_i = \langle x, y \rangle = \langle Ax, Ay \rangle = \overline{a_{lj}} a_{lk} \overline{x_j} y_k$$

Taking $x = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ yields $y_1 = \overline{a_{l1}} a_{lk} y_k$ which is satisfied for arbitrary y if and only if $\overline{a_{l1}} a_{lk} = \delta_{1k}$. The argument can be repeated with the position of the 1 in x transferred, such that $\overline{a_{lj}} a_{lk} = \delta_{jk}$ and hence A has orthonormal column vectors.

Conversely, suppose the column vectors of A are orthonormal. Then

$$\langle Ax, Ay \rangle = \overline{a_{lj}} a_{lk} \overline{x_j} y_k = \delta_{jk} \overline{x_j} y_k = \overline{x_i} y_i = \langle x, y \rangle$$

as desired. Now take some $n \times n$ matrix B. We have

$$\langle Bx, y \rangle = \overline{b_{kl}} \overline{x_l} y_k = \overline{b_{lk}} \overline{x_k} y_l = \langle x, B^* y \rangle$$

as desired. Hence, if a matrix A preserves the inner product,

$$\langle x, y \rangle = \langle Ax, Ay \rangle = \langle x, A^* A y \rangle$$

We can see this is satisfied only if $A^* A y = y$ or equivalently $A^* A = I$, since if there were an index in which y and $A^* A y$ differ, we can construct a vector x which vanishes everywhere except that index and derive a contradiction. ■

4. *Generalized orthogonal groups.* Let $[\cdot, \cdot]_{n,k}$ be the symmetric bilinear form on $\mathbb{R}^{n+k}$. Let g be defined as

$$g = \begin{pmatrix} I_n & 0 \\ 0 & -I_K \end{pmatrix}.$$

Show that for all $x, y \in \mathbb{R}^{n+k}$,

$$[x, y]_{n,k} = \langle x, gy \rangle$$

Show that a $(n+k) \times (n+k)$ real matrix A is in $\mathsf{O}(n;k)$ if and only if $A^{\mathrm{T}} g A = g$. Show that $\mathsf{O}(n;k)$ and $\mathsf{SO}(n;k)$ are subgroups of $\mathsf{GL}(n+k, \mathbb{R})$ and are matrix lie groups.

Proof. By direct computation,

$$[x, y]_{n,k} = x_1 y_1 + \cdots + x_n y_n - x_{n+1} y_{n+1} - \cdots - x_{n+k} y_{n+k} = \langle x, gy \rangle$$

A matrix in $\mathsf{O}(n;k)$ satisfies the following properties:

$$\begin{aligned} &1. [A^{(l)}, A^{(j)}]_{n,k} = [a_{il}, a_{ij}]_{n,k} = 0 && l \neq j, \\ &2. [A^{(l)}, A^{(l)}]_{n,k} = [a_{il}, a_{il}]_{n,k} = 1 && 1 \leq l \leq n, \\ &3. [A^{(l)}, A^{(l)}]_{n,k} = [a_{il}, a_{il}]_{n,k} = -1 && n+1 \leq l \leq n+k, \end{aligned}$$

Take some real matrix A. By direct computation, we can form the product

$$A^{\mathrm{T}}gA = \begin{pmatrix} a_{11} & \cdots & a_{n+k\,1} \\ \vdots & \ddots & \vdots \\ a_{1\,n+k} & \cdots & a_{n+k\,n+k} \end{pmatrix} \begin{pmatrix} I_n & 0 \\ 0 & -I_K \end{pmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1\,n+k} \\ \vdots & \ddots & \vdots \\ a_{n+k\,1} & \cdots & a_{n+k\,n+k} \end{pmatrix}$$
$$= \begin{pmatrix} [a_{i1}, a_{i1}]_{n,k} & [a_{i1}, a_{i2}]_{n,k} & \cdots & [a_{i1}, a_{i\,n+k}]_{n,k} \\ [a_{i2}, a_{i1}]_{n,k} & [a_{i2}, a_{i2}]_{n,k} & \cdots & [a_{i1}, a_{i\,n+k}]_{n,k} \\ \vdots & \vdots & \ddots & \vdots \\ [a_{i\,n+k}, a_{i1}]_{n,k} & [a_{i\,n+k}, a_{i2}]_{n,k} & \cdots & [a_{i\,n+k}, a_{i\,n+k}]_{n,k} \end{pmatrix}$$

Suppose $A^{\mathrm{T}}gA = g$. It can thus be observed from the explicit form of $A^{\mathrm{T}}gA$ that the symmetric bilinear form of two distinct columns (condition 1), corresponding to off-diagonal elements, is always 0, as is necessary for membership in $\mathsf{O}(n;k)$. Moreover, the first n diagonal elements correspond exactly to the requirements of condition 2, and the remaining diagonal elements correspond to condition 3, so that $A \in \mathsf{O}(n;k)$.

Conversely, suppose $A \in \mathsf{O}(n;k)$. We have already determined the explicit form of $A^{\mathrm{T}}gA$ and it will be observed that we can directly input the necessary conditions into the matrix such that $a^{\mathrm{T}}gA = g$, thus establishing the desired equivalence.

For $A \in \mathsf{O}(n;k)$, we have $\det(g) = \det(A^{\mathrm{T}}gA) = \det(A^{\mathrm{T}})\det(g)\det(A)$. Since $\det(g) = \pm 1$, we have $\det(A) = \det(A^{\mathrm{T}}) \neq 0$ and hence $A \in \mathsf{GL}(n+k, \mathbb{R})$. To show that $\mathsf{O}(n;k)$ is a subgroup of $\mathsf{GL}(n+k, \mathbb{R})$, take $A, B \in \mathsf{O}(n;k)$. We inherit associativity from the group operation of matrix multiplication of $\mathsf{GL}(n+k, \mathbb{R})$. We have $I \in \mathsf{O}(n;k)$ since clearly $I^{\mathrm{T}}gI = g$. We have $(AB)^{\mathrm{T}}gAB = B^{\mathrm{T}}A^{\mathrm{T}}gAB = B^{\mathrm{T}}(A^{\mathrm{T}}gA)B = B^{\mathrm{T}}gB = g$ so that $AB \in \mathsf{O}(n;k)$ and we hence have closure. Finally, given the commutativity of inversion and the transpose, and noting that $A^{-1} \in \mathsf{GL}(n+k, \mathbb{R})$ by definition of the group so that we may safely write down A^{-1} as a matrix, we have

$$A^{\mathrm{T}}gA = g \implies A^{\mathrm{T}}g = gA^{-1} \implies g = (A^{\mathrm{T}})^{-1}gA^{-1} = (A^{-1})^{\mathrm{T}}gA^{-1}$$

so that $A^{-1} \in \mathsf{O}(n;k)$. This yields $\mathsf{O}(n;k)$ as a subgroup of $\mathsf{GL}(n+k, \mathbb{R})$. To show that it is a Lie group, take some sequence $A_m \to A$, where $A_m \in \mathsf{O}(n;k)$. Consider the sequence

$$A_m^{\mathrm{T}}gA_m = \begin{pmatrix} [a_{i1}^m, a_{i1}^m]_{n,k} & [a_{i1}^m, a_{i2}^m]_{n,k} & \cdots & [a_{i1}^m, a_{i\,n+k}^m]_{n,k} \\ [a_{i2}^m, a_{i1}]_{n,k}^m & [a_{i2}^m, a_{i2}]_{n,k}^m & \cdots & [a_{i1}^m, a_{i\,n+k}]_{n,k}^m \\ \vdots & \vdots & \ddots & \vdots \\ [a_{i\,n+k}^m, a_{i1}^m]_{n,k} & [a_{i\,n+k}^m, a_{i2}^m]_{n,k} & \cdots & [a_{i\,n+k}^m, a_{i\,n+k}^m]_{n,k} \end{pmatrix}$$

where m denotes the the usage of A_m. Consider for instance $[a_{i1}^m, a_{i1}^m]_{n,k} = a_{11}^m a_{11}^m + a_{22}^m a_{22}^m + \cdots + a_{nn}^m a_{nn}^m - a_{n+1\,n+1}^m - \cdots - a_{n+k\,n+k}^m$. By the convergence of A_m, we have that $a_{11}^m \to a_{11}$ where a_{11} is the upper left entry in the matrix A. By the Algebraic Limit Theorem, we have $a_{11}^m a_{11}^m \to a_{11}a_{11}$ and similarly for the other products. Fix $\epsilon > 0$. For each matrix entry a_{jl}^m in A_m, and for $i \in \{1, \ldots, n+k\}$, assign an $N \in \mathbb{Z}^+$ such that $m \geq N$ implies $|a_{ij}^m a_{il}^m - a_{ij}| < \epsilon/(n+k)$. Take N as the maximal such N. Hence, for a fixed $j, l \in \{1, \ldots, n+k\}$, and with $m \geq N$, we have

$$
\begin{aligned}
&\left|[a^m_{ij}, a^m_{il}]_{n,k} - [a_{ij}, a_{il}]_{n,k}\right| \\
=&\Big|a^m_{1j}a^m_{1l} + \ldots a^m_{nj}a^m_{nl} - a^m_{n+1\,j}a^m_{n+1\,l} - \cdots - a^m_{n+j\,j}a^m_{n+j\,l} \\
&\quad - \Big(a_{1j}a_{1l} + \ldots a_{nj}a_{nl} - a_{n+1\,j}a_{n+1\,l} - \cdots - a_{n+j\,j}a_{n+j\,l}\Big)\Big| \\
\leq&\left|a^m_{1j}a^m_{1l} - a_{1j}a_{1l}\right| + \cdots + \left|a^m_{n+k\,j}a^m_{n+k\,l} - a_{n+k\,j}a_{n+k\,l}\right| \\
&< \frac{\epsilon}{n+k} + \cdots + \frac{\epsilon}{n+k} = (n+k)\frac{\epsilon}{n+k} = \epsilon
\end{aligned}
$$

so that product $A^{\mathrm{T}}_m g A_m$ converges element-wise to g. Thus, $A^{\mathrm{T}} g A = g$ and so $A \in \mathsf{O}(n;k)$ such that $\mathsf{O}(n;k)$ is a Lie group.

Now take some $A, B \in \mathsf{SO}(n;k)$. We have $\mathsf{SO}(n;k) \subseteq \mathsf{GL}(n+k,\mathbb{R})$ once again. Associativity is again inherited, and $\det(I) = 1$ so that $I \in \mathsf{SO}(n;k)$. The exact considerations above yield $AB \in O(n;k)$ and we have $\det(AB) = \det(A)\det(B) = 1$ so that $AB \in \mathsf{SO}(n;k)$. Finally, $1 = \det(I) = \det(A^{-1}A) = \det(A^{-1}\det(A) = \det(A^{-1})$ gives $A^{-1} \in \mathsf{SO}(n;k)$ so that $\mathsf{SO}(n;k)$ forms a subgroup of $\mathsf{GL}(n+k,\mathbb{R})$ as well. From the previous argument we have that any subsequence $A_m \to A$ with $A_m \in \mathsf{SO}(n;k)$ must have $A \in \mathsf{SO}(n;k)$, and from the continuity of the determinant we must have $\det(a) = 1$ and so $A \in \mathsf{SO}(n;k)$. This shows that $\mathsf{SO}(n;k)$ is a Lie group as well. ■

6. *The groups* $\mathsf{O}(2)$ *and* $\mathsf{SO}(2)$. Show that the matrix

$$
\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}
$$

is in $\mathsf{SO}(2)$ and that

$$
\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} = \begin{pmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{pmatrix}.
$$

Show that every element A of $\mathsf{O}(2)$ is of one of two forms:

$$
\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.
$$

Proof. If a matrix A can written as

$$
A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}
$$

with $a^2_{11} + a^2_{12} = 1$, $a^2_{21} + a^2_{22} = 1$, and $a_{11}a_{12} + a_{21}a_{22} = 0$, as well as $\det(A) = 0$, then $A \in \mathsf{SO}(2)$. We can immediately compute $a^2_{11} + a^2_{12} = \cos^2\theta + \sin^2\theta = 1 = a^2_{21} + a^2_{22}$ and that $a_{11}a_{12} + a_{21}a_{22} = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$. We have $\det(A) = \cos^2\theta + \sin^2\theta = 1$ and hence the matrix is in $\mathsf{SO}(2)$.

We can compute the product

$$\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\cos\phi & -\sin\phi \\ \sin\phi & \cos\phi\end{pmatrix} = \begin{pmatrix}\cos\theta\cos\phi - \sin\theta\sin\phi & -\cos\theta\sin\phi - \sin\theta\cos\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & -\sin\theta\sin\phi + \cos\theta\cos\phi\end{pmatrix}$$
$$= \begin{pmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi)\end{pmatrix}$$

by the standard trigonometric sum identities. Now take some $A \in \mathsf{O}(2)$. Since $|a_{11}|, |a_{22}| < 1$, we may label them as $\cos\theta_1$ and $\cos\theta_2$, respectively, for some $\theta_1, \theta_2 \in [0, 2\pi)$; this is certainly not the only way to label these elements, but it is a valid choice given the norm of the sin and cos functions. The normality of the column vectors requires A to be of the form

$$A = \begin{pmatrix}\cos\theta_1 & \pm\sin\theta_2 \\ \pm\sin\theta_1 & \cos\theta_2\end{pmatrix}$$

by the standard trigonometric identity $\sin^2\theta + \cos^2\theta = 1$. We now investigate the various cases for the parity of the sin term using the last condition that the sum of the rows must vanish.

If we take the positive form of both of the sin arguments, we have $\sin(\theta_1+\theta_2) = \cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2 = 0$, which is satisfied if and only if $\theta_1 = -\theta_2 = \theta$ or $\theta_1 = \pi - \theta_2 = \theta$ given our choice of domain. In the first case, we can write

$$A = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix},$$

and in the second case we can write

$$A = \begin{pmatrix}\cos\theta & \sin\theta \\ \sin\theta & -\cos\theta\end{pmatrix},$$

both of which are of the desired form. If we now take the negative form on the $\sin\theta_1$ argument, we have $\sin(\theta_2 - \theta_1) = \cos\theta_1\sin\theta_2 - \sin\theta_1\cos\theta_2 = 0$, which is satisfied if and only if $\theta_1 = \theta_2 = \theta$ or $\theta_1 = \theta_2 - \pi = \theta$. In the first case,

$$A = \begin{pmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{pmatrix} = \begin{pmatrix}\cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta)\end{pmatrix} \xrightarrow[\phi:=-\theta]{} \begin{pmatrix}\cos\phi & -\sin\phi \\ \sin\phi & \cos\phi\end{pmatrix},$$

where we have defined ϕ as $-\theta$; based on the behavior of the sin and cos functions, the parity of the elements are now such that the entires of A are consistent with our new definition of θ, and the three conditions for orthogonality are thus automatically satisfied. The form of the matrix is still consistent with the case we set out to investigate on θ; our definition of ϕ serves as an alternative characterization which is of the desired form. Analogously, taking $\theta_1 = \theta_2 - \pi = \theta$ gives

$$A = \begin{pmatrix}\cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta\end{pmatrix} = \begin{pmatrix}\cos(-\theta) & \sin(-\theta) \\ \sin(-\theta) & -\cos(-\theta)\end{pmatrix} \xrightarrow[\phi:=-\theta]{} \begin{pmatrix}\cos\phi & \sin\phi \\ \sin\phi & -\cos\phi\end{pmatrix},$$

again of the desired form. Now take the positive form of $\sin\theta_1$ and the negative of $\sin\theta_2$. The condition is once again $\sin(\theta_1 - \theta_2) = 0$, which can be characterized by $\theta_1 = \theta_2 = \theta$ or $\theta_1 = \theta_2 - \pi = \theta$. In the first case, we have

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

and in the second case we have

$$A = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{pmatrix},$$

both of which are of the desired form. Finally, take the negative argument on both $\sin\theta_1$ and $\sin\theta_2$. The relevant condition is then $-\sin(\theta_1 + \theta_2) = -[\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2] = 0$, giving $\theta_2 = -\theta_1 = \theta$ or $\theta_1 = \pi - \theta_2 = \theta$. In the first case,

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and in the second case we have

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} = \begin{pmatrix} \cos(-\theta) & \sin(-\theta) \\ \sin(-\theta) & -\cos(-\theta) \end{pmatrix} \xrightarrow[\phi := -\theta]{} \begin{pmatrix} \cos\phi & \sin\phi \\ \sin\phi & -\cos\phi \end{pmatrix}.$$

This exhausts all possible cases and hence $A \in \mathsf{O}(n)$ can always be written in one of the two specified forms. ■

10. *The Heisenberg group.* Determine the center $Z(H)$ of the Heisenberg group H. Show that the quotient group $H/Z(H)$ is abelian.

Proof. Fix some $A \in H$ given as

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \quad a, b, c \in \mathbb{R}.$$

Take an arbitrary $B \in H$ given as

$$B = \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \quad a', b', c' \in \mathbb{R}.$$

We want to find the condition which a, b, and c satisfy such that $AB = BA$ for all $B \in H$. We compute

$$AB = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a + a' & b + b' + ac' \\ 0 & 1 & c' + c' \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$BA = \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b+b'+a'c \\ 0 & 1 & c'+c' \\ 0 & 0 & 1 \end{pmatrix}.$$

We therefore have equivalence if and only if $ac' = ca'$ for all $a', c' \in \mathbb{R}$. It can be readily shown by taking $(a', c') = (1, 0)$ and $(a', c') = (0, 1)$ that the only way for this to happen is for $a = 0 = c$. Hence, we can describe the center $Z(H)$ as

$$Z(H) = \left\{ \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

We can formulate the equivalence classes under $Z(H)$ by finding the elements $A, B \in H$ such that $AB^{-1} \in Z(H)$. Filling in the upper right entries for A as a, b, and c and for B as a', b', c', we use the explicit form of the inverse matrix presented in the text to find

$$AB^{-1} = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a' & a'c' - b' \\ 0 & 1 & -c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a-a' & a'c' - b' - ac' + b \\ 0 & 1 & c - c' \\ 0 & 0 & 1 \end{pmatrix}$$

We hence have $AB^{-1} \in Z(G)$ if and only if $a = a'$ and $c = c'$. The equivalence classes can hence be labeled as $[a, c]$. It is easy to check that the group operation $[a, c][a', c'] = [a + a', c + c'] = [a', c'][a, c]$ is commutative and hence $H/Z(H)$ is abelian. ■

2 Lie Algebras and the Exponential Mapping

1. The Schwarz inequality from elementary analysis tells us that for $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$ in $\mathbb{C}^n$, we have

$$|u_1 v_1 + \cdots + u_n v_n|^2 \leq \left(\sum_{k=1}^{n} |u_k|^2 \right) \left(\sum_{k=1}^{n} |v_k|^2 \right).$$

Use this to verify that $\|XY\| \leq \|X\|\|Y\|$ for all $X, Y \in M_n(\mathbb{C})$, where the norm $\|X\|$ of a matrix X is as defined in the text.

Proof. Take arbitrary $X, Y \in M_n(\mathbb{C})$. Applying the Schwarz inequality term-by-term in the matrix $\|XY\|$, we have

$$\|XY\|^2 = \sum_{k,l=1}^{n} \left| \sum_{i=1}^{n} X_{ki} Y_{il} \right|^2 \leq \sum_{k,l=1}^{n} \left(\sum_{i=1}^{n} |X_{ki}|^2 \sum_{j=1}^{n} |Y_{jl}|^2 \right) = \sum_{i,j,k,l=1}^{n} |X_{ki}|^2 |Y_{jl}|^2.$$

We will show that the right hand side is equivalent to $\|X\|^2 \|Y\|^2$. We compute

$$\begin{aligned} \|X\|^2 \|Y\|^2 &= \left(\sum_{k,i=1}^{n} |X_{ki}|^2 \right) \left(\sum_{l,j=1}^{n} |Y_{jl}|^2 \right) \\ &= \left(\sum_{k=1}^{n} \sum_{i=1}^{n} |X_{ki}|^2 \right) \left(\sum_{l=1}^{n} \sum_{j=1}^{n} |Y_{jl}|^2 \right) \\ &= \sum_{i,j,k,l=1}^{n} |X_{ki}|^2 |Y_{jl}|^2, \end{aligned}$$

where we have employed the commutation of summations. Since the norm is a non-negative real number, we hence have $\|XY\| \leq \|X\|\|Y\|$ for all $X, Y \in M_n(\mathbb{C})$. ■

2. Show that for $X \in M_n(\mathbb{C})$ and any orthonormal basis $\{u_1, \ldots, u_n\}$ of $\mathbb{C}^n$, $\|X\|^2 = \sum_{j,k=1}^{n} |\langle u_j, X u_k \rangle|^2$. Now show that if v is an eigenvector of X with eigenvalue λ, then $|\lambda| \leq \|X\|$.

Proof. Given an arbitrary orthonormal basis $u := \{u_1, \ldots, u_n\}$, a matrix $X \in M_n(\mathbb{C})$ can always be expressed as

$$X \xrightarrow[u \text{ basis}]{} \begin{pmatrix} \langle u_1, X u_1 \rangle & \cdots & \langle u_1, X u_n \rangle \\ \vdots & \ddots & \vdots \\ \langle u_n, X u_1 \rangle & \cdots & \langle u_n, X u_n \rangle \end{pmatrix}.$$

This can be explicitly verified by decomposing an arbitrary vector $v \in \mathbb{C}^n$ into the basis u and showing that application of the matrix achieves the desired linear transformation on each of the vectors comprising the basis; this works because the inner product is basis-invariant. The desired norm is then found directly by summing the squares of the entries.

Now take some eigenvector v with eigenvalue λ. Since u was arbitrary, take $u_1 = v/|v|$ and construct an orthonormal basis which follows; this could be done, for instance, by rotating the standard basis in N dimensional space until u_1 coincides with v. Then we have $v = \begin{pmatrix} v_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ and

$$Xv = \begin{pmatrix} v_0\langle u_1, Xu_1\rangle \\ v_0\langle u_2, Xu_1\rangle \\ \vdots \\ v_0\langle u_n, Xu_1\rangle \end{pmatrix} = \begin{pmatrix} \lambda v_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \lambda v,$$

yielding $\lambda = \langle u_1, Xu_1\rangle$. Hence,

$$|\lambda| = |\langle u_1, Xu_1\rangle| \leq \left(\sum_{j,k=1}^{n} |\langle u_j, Xu_k\rangle|^2 \right)^{1/2} = \|X\|$$

as desired. ■

3. *The product rule.* Recall that a matrix-valued function $A(t)$ is said to be smooth if each A_{ij} is smooth. The derivative of such a function is defined as

$$\left(\frac{\mathrm{d}A}{\mathrm{d}t}\right)_{ij} = \frac{\mathrm{d}A_{ij}}{\mathrm{d}t}$$

or, equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}A(t) = \lim_{h\to 0} \frac{A(t+h) - A(t)}{h}.$$

Let $A(t)$ and $B(t)$ be two such smooth functions. Prove that $A(t)B(t)$ is again smooth and that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[A(t)B(t)\right] = \frac{\mathrm{d}A}{\mathrm{d}t}B(t) + A(t)\frac{\mathrm{d}B}{\mathrm{d}t}.$$

Proof. With the matrices

$$A(t) = \begin{pmatrix} a_{11}(t) & \cdots & a_{1n}(t) \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \cdots & a_{nn}(t) \end{pmatrix}, \qquad B(t) = \begin{pmatrix} b_{11}(t) & \cdots & b_{1n}(t) \\ \vdots & \ddots & \vdots \\ b_{n1}(t) & \cdots & b_{nn}(t) \end{pmatrix},$$

we form the matrix product

$$A(t)B(t) = \begin{pmatrix} a_{1i}(t)b_{i1}(t) & \cdots & a_{1i}(t)b_{in}(t) \\ \vdots & \ddots & \vdots \\ a_{ni}(t)b_{i1}(t) & \cdots & a_{ni}(t)b_{in}(t) \end{pmatrix},$$

where the i is summed over $\{1, \ldots, n\}$. It is readily observed that entry is a sum of products of smooth functions and hence is continuous itself, thereby implying that $A(t)B(t)$ is smooth itself. To find the explicit form, consider some $a_{ji}(t)b_{ik}(t)$ for some fixed $j, k \in \{1, \ldots, n\}$; this is equivalent to taking a element of the matrix $A(t)B(t)$. We now consider the time derivative of this element. We compute by the product rule for complex valued functions

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[a_{ji}(t)b_{ik}(t)\right] = \frac{\mathrm{d}a_{ji}(t)}{\mathrm{d}t}b_{ik}(t) + a_{ji}(t)\frac{\mathrm{d}b_{ik}(t)}{\mathrm{d}t},$$

so that by the definition of the matrix derivative,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[(A(t)B(t)\right] = \begin{pmatrix} \frac{\mathrm{d}a_{1i}(t)}{\mathrm{d}t}b_{i1}(t) + a_{1i}(t)\frac{\mathrm{d}b_{i1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{1i}(t)}{\mathrm{d}t}b_{in}(t) + a_{1i}(t)\frac{\mathrm{d}b_{in}(t)}{\mathrm{d}t} \\ \vdots & \ddots & \vdots \\ \frac{\mathrm{d}a_{ni}(t)}{\mathrm{d}t}b_{i1}(t) + a_{ni}(t)\frac{\mathrm{d}b_{i1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{ni}(t)}{\mathrm{d}t}b_{in}(t) + a_{ni}(t)\frac{\mathrm{d}b_{in}(t)}{\mathrm{d}t} \end{pmatrix}$$

But now considering the derivatives

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \begin{pmatrix} \frac{\mathrm{d}a_{11}}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{1n}}{\mathrm{d}t} \\ \vdots & \ddots & \vdots \\ \frac{\mathrm{d}a_{n1}}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{nn}}{\mathrm{d}t} \end{pmatrix}, \qquad \frac{\mathrm{d}B}{\mathrm{d}t} = \begin{pmatrix} \frac{\mathrm{d}b_{11}}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}b_{1n}}{\mathrm{d}t} \\ \vdots & \ddots & \vdots \\ \frac{\mathrm{d}b_{n1}}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}b_{nn}}{\mathrm{d}t} \end{pmatrix},$$

we have

$$\frac{\mathrm{d}A(t)}{\mathrm{d}t}B(t) + A(t)\frac{\mathrm{d}B(t)}{\mathrm{d}t} = \begin{pmatrix} \frac{\mathrm{d}a_{1i}(t)}{\mathrm{d}t}b_{i1}(t) + a_{1i}(t)\frac{\mathrm{d}b_{i1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{1i}(t)}{\mathrm{d}t}b_{in}(t) + a_{1i}(t)\frac{\mathrm{d}b_{in}(t)}{\mathrm{d}t} \\ \vdots & \ddots & \vdots \\ \frac{\mathrm{d}a_{ni}(t)}{\mathrm{d}t}b_{i1}(t) + a_{ni}(t)\frac{\mathrm{d}b_{i1}(t)}{\mathrm{d}t} & \cdots & \frac{\mathrm{d}a_{ni}(t)}{\mathrm{d}t}b_{in}(t) + a_{ni}(t)\frac{\mathrm{d}b_{in}(t)}{\mathrm{d}t} \end{pmatrix} = \frac{\mathrm{d}}{\mathrm{d}t}\left[(A(t)B(t)\right]$$

as was desired. ■

4. Show that for all $X \in M_n(\mathbb{C})$,

$$\lim_{m\to\infty}\left[I + \frac{X}{m}\right]^m = e^X.$$

Proof. We note that our proof is surely not the easiest way to prove this result. Since I and X and all combinations therein commute, we can for any finite m invoke the binomial expansion

$$\begin{aligned}\left[I+\frac{X}{m}\right]^m &= I+\binom{m}{1}\frac{X}{m}+\binom{m}{2}\left(\frac{X}{m}\right)^2+\cdots+\binom{m}{m}\left(\frac{X}{m}\right)^2 \\ &= I+X+\frac{1}{2!}\frac{m-1}{m}X^2+\cdots+\frac{1}{m!}\frac{(m-1)(m-2)\cdots(2)(1)}{m^m}X^m.\end{aligned}$$

A comparison with the exponential function yields

$$\begin{aligned}e^X-\left[I+\frac{X}{m}\right]^m &= \frac{1}{2!}\left(1-\frac{m-1}{m}\right)X^2+\frac{1}{3!}\left(1-\frac{(m-1)(m-2)}{m(m-1)}\right)X^3 \\ &\quad+\cdots+\frac{1}{m!}\left(1-\frac{(m-1)!}{m^{m-1}}\right)X^m+\frac{1}{(m+1)!}X^{m+1}+\ldots\end{aligned}$$

and our goal is to show that

$$\lim_{m\to\infty}\left\|e^X-\left[I+\frac{X}{M}\right]^m\right\|=0.$$

Take $\|e^X\|=s$ for some $s\in\mathbb{R}$, with $s\geq 0$. It follows that

$$\begin{aligned}\lim_{m\to\infty}\left\|e^X-\left[I+\frac{X}{M}\right]^m\right\| &= \left\|\frac{1}{2!}\left(1-\frac{m-1}{m}\right)X^2+\frac{1}{3!}\left(1-\frac{(m-1)(m-2)}{m(m-1)}\right)X^3\right. \\ &\quad\left.+\cdots+\frac{1}{m!}\left(1-\frac{(m-2)(m-1)\cdots(1)}{m(m-1)\cdots(2)}\right)X^m+\frac{1}{(m+1)!}X^{m+1}+\cdots\right\| \\ &\leq \left\|\frac{1}{2!}\left(1-\frac{m-1}{m}\right)X^2\right\|+\left\|\frac{1}{3!}\left(1-\frac{(m-1)(m-2)}{m(m-1)}\right)X^3\right\| \\ &\quad+\cdots+\left\|\frac{1}{m!}\left(\frac{(m-2)(m-1)\cdots(1)}{m(m-1)\cdots(2)}\right)X^m\right\|+\left\|\frac{1}{(m+1)!}X^{m+1}\right\|+\cdots \\ &\leq \frac{s^2}{2!}\left(1-\frac{m-1}{m}\right)+\frac{s^3}{3!}\left(1-\frac{(m-1)(m-2)}{m(m-1)}\right) \\ &\quad+\cdots+\frac{s^m}{m!}\left(\frac{(m-1)(m-2)\cdots(1)}{m(m-1)\cdots(2)}\right)+\frac{s^{m+1}}{(m+1)!}+\cdots\end{aligned}$$

Fix $\epsilon>0$. We will find M such that $m\geq M$ implies that the above summation is less than ϵ. First consider the terms up to (and including) the $\frac{s^m}{m!}$ term and denote the sequence by a_n, where n corresponds to the factorial and hence $n\in\{2,\ldots,m\}$. Consider the ratio of two consecutive terms a_n and a_{n+1}. We compute

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{\frac{s^{n+1}}{(n+1)!}\left(1-\frac{(m-1)(m-2)\ldots(m-n)}{m(m-1)\cdots(m-n+1)}\right)}{\frac{s^n}{n!}\left(1-\frac{(m-1)(m-2)\ldots(m-n+1)}{m(m-1)\cdots(m-n+2)}\right)} \\ &= \frac{s}{n}\frac{m(m-1)\cdots(m-n+1)-(m-1)(m-2)\cdots(m-n)}{m(m-1)\cdots(m-n+1)-(m-1)(m-2)\cdots(m-n+1)(m-n+1)} \\ &= \frac{s}{n}\frac{m-(m-n)}{m-(m-n+1)} = \frac{s}{n}\frac{n}{n-1} = \frac{s}{n-1}\end{aligned}$$

Hence, if we take $m > \frac{s^2e^s}{\epsilon}$,

$$\sum_{n=2}^{m} a_n = \frac{s^2}{2m} + \frac{s^3}{2m} + \frac{s^4}{4m} + \frac{s^5}{12m} + \cdots = \frac{s^2}{2m}\sum_{n=0}^{m-2}\frac{s^n}{n!} < \frac{s^2e^s}{2m} < \frac{\epsilon}{2}.$$

Next, we consider the sequence $b_n = \sum_{n=m+1}^{\infty}\frac{s^n}{n!}$. We will assume without loss of generality that $\epsilon < 1$. Noting that $\lim n \to \infty \frac{s^n}{n!} = 0$, if we take m such that $\frac{s^{m+1}}{(m+1)!} < \frac{\epsilon}{2}(1-\epsilon)$ and $\frac{s}{m+2} < \epsilon$, we can write

$$\begin{aligned}\sum_{n=m+1}^{\infty} b_n &= \frac{s^{m+1}}{(m+1)!} + \frac{s^{m+2}}{(m+2)!} + \cdots \leq \frac{s^{m+1}}{(m+1)!} + \frac{s^{m+1}}{(m+1)!}\left(\frac{s}{m+2}\right) + \frac{s^{m+1}}{(m+1)!}\left(\frac{s}{m+2}\right)^2 + \cdots \\ &< \frac{s^{m+1}}{(m+1)!}\left[1+\epsilon+\epsilon^2+\cdots\right] < \frac{\epsilon(1-\epsilon)}{2}\left[1+\epsilon+\epsilon^2+\cdots\right] < \frac{\epsilon}{2}\end{aligned}$$

as the sum of the geometric series. We finally conclude that given a fixed $\epsilon < 0$, for $m > \max\left(\frac{s^2e^s}{\epsilon}, \frac{s}{\epsilon} - 2\right)$ and which satisfies $\frac{s^{m+1}}{(m+1)!} < \frac{\epsilon}{2}(1-\epsilon)$,

$$\begin{aligned}\left|\left|\left|e^X - \left[I+\frac{X}{M}\right]^m\right|\right|\right| &= \left\|e^X - \left[I+\frac{X}{M}\right]^m\right\| \\ &\leq \frac{s^2}{2!}\left(1-\frac{m-1}{m}\right) + \frac{s^3}{3!}\left(1-\frac{(m-1)(m-2)}{m(m-1)}\right) \\ &\quad + \cdots + \frac{s^m}{m!}\left(\frac{(m-1)(m-2)\cdots(1)}{m(m-1)\cdots(2)}\right) + \frac{s^{m+1}}{(m+1)!} + \cdots \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} < \epsilon,\end{aligned}$$

so that

$$\lim_{m\to\infty}\left\|e^X - \left[I+\frac{X}{M}\right]^m\right\| = 0.$$

and hence

$$\lim_{m\to\infty}\left[I+\frac{X}{M}\right]^m = e^X$$

as was desired. ■

5. Using Theorem B.7, show that every $n \times n$ complex matrix A is the limit of a sequence of diagonalizable matrices.

Proof. Denote the eigenvalues of A as $\lambda_1, \ldots, \lambda_k$, where in general $k \leq n$ covers multiplicities. By Theorem B.7, we can write $A = CBC^{-1}$, where $C \in \mathsf{GL}(n, \mathbb{C})$ and B is upper triangular. The diagonal elements of B are its eigenvalues, by the characteristic polynomial.

If they are all distinct, then B is diagonalizable as $B = DED^{-1}$ and hence $A = CBC^{-1} = CDED^{-1}E^{-1} = (CD)E(DC)^{-1}$ implies that A is itself diagonalizable and we can simply take the sequence $A, A, \ldots$ which obviously converges to A.

If there are multiplicities, let the smallest gap between distinct eigenvalues be δ. Define the upper triangular matrix B_m as follows. For each eigenvalue λ_i with a multiplicity of 1, add a corresponding entry λ_i along the diagonal. For each eigenvalue λ_i with multiplicity of $l > 1$, add entries $\lambda_i + \frac{j\delta}{2lm}$ for $j \in \{1, \ldots, l\}$. Leave the elements above the main diagonal as in B. Then each B_m is an upper triangular matrix with distinct eigenvalues and hence diagonalizable; by the logic above, $A_m = CB_mC^{-1}$ is diagonalizable as well. Since $\lim_{m\to\infty} B_m = B$, it follows that $\lim_{m\to\infty} A_m = A$ and we are done. ■

6. Show that every 2×2 matrix X with trace zero satisfies

$$X^2 = -\det(X) I.$$

If X is 2×2 with trace zero, show by direct calculation using the power series that

$$e^X = \cos\left(\sqrt{\det X}\right) I + \frac{\sin\left(\sqrt{\det X}\right)}{\sqrt{\det X}} X.$$

Use this to give an alternate derivation of the result presented in the text that

$$X = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix} \implies e^X = \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix}.$$

Proof. An arbitrary matrix X with trace zero can be expressed as

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

for some $a, b, c \in \mathbb{C}$. We compute

$$X^2 = \begin{pmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{pmatrix} = (a^2+bc)I = -\det(X)I.$$

Denote $\lambda := \det(X)$. We have for $m = 2k, k \in \mathbb{Z}^+$ that $X^m = (-\lambda)^k I$ and for $m = 2k+1, k \in \mathbb{Z}^+$ that $X^m = (-\lambda)^k X$. Note that the square root $\sqrt{\lambda}$ is well defined on the principal branch $-\pi < \theta < \pi$ via the mapping $re^{i\theta} \to \sqrt{r}e^{i\theta/2}$ where the argument are assumed to be fixed. Denote $\phi := \sqrt{\lambda}$. The exponential series expansion yields

$$\begin{aligned} e^X &= \sum_{m=0}^{\infty} \frac{X^m}{m!} = I + X - \frac{I}{2!}\lambda - \frac{X}{3!}\lambda + \frac{I}{4!}\lambda^2 + \cdots \\ &= \left(1 - \frac{\lambda}{2!} + \frac{\lambda^2}{4!} + \cdots\right) I + \left(1 - \frac{1}{3!}\lambda + \frac{1}{5!}\lambda^2 + \cdots\right) X \\ &= \left(1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} + \cdots\right) I + \frac{1}{\phi}\left(\frac{\phi}{1!} - \frac{\phi^3}{3!} + \frac{\phi^5}{5!} + \cdots\right) X \\ &= \cos(\phi) I + \frac{\sin(\phi)}{\phi} X \\ &= \cos\left(\sqrt{\det X}\right) I + \frac{\sin\left(\sqrt{\det X}\right)}{\sqrt{\det X}} X, \end{aligned}$$

by the complex series expansions of sin and cos. It follows then for the specified matrix that, if $a = re^{i\theta}, \theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, then $\sqrt{a^2} = a$ and hence

$$\begin{aligned} e^X &= \cos\left(\sqrt{\det X}\right) I + \frac{\sin\left(\sqrt{\det X}\right)}{\sqrt{\det X}} X \\ &= \cos(a)\, I + \frac{\sin(a)}{a} X \\ &= \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix}. \end{aligned}$$

If $a = re^{i\theta}, \theta \in (\frac{\pi}{2}, \frac{3\pi}{2})$, then $\sqrt{a^2} = -a$ and hence

$$\begin{aligned} e^X &= \cos\left(\sqrt{\det X}\right) I + \frac{\sin\left(\sqrt{\det X}\right)}{\sqrt{\det X}} X \\ &= \cos(-a)\, I + \frac{\sin(-a)}{-a} X \\ &= \cos(a)\, I + \frac{\sin(a)}{a} X \\ &= \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix}. \end{aligned}$$

Inspection of the boundaries completes the derivation. ■

7. Use the result of Exercise 6 to compute the exponential of the matrix

$$X = \begin{pmatrix} 4 & 3 \\ -1 & 2 \end{pmatrix}.$$

Proof. First we show the following lemma, which was stated but not proven in the text.

Lemma. *Given arbitrary $X, Y \in M_n(\mathbb{C})$, if X and Y commute then $e^{X+Y} = e^X e^Y$.*

Proof. Consider the product $(X+Y)^n$ for $n \in \mathbb{Z}^+$. By commutation, we can write $(X+Y)^2 = X^2 + XY + YX + Y^2 = X^2 + 2XY + Y^2$, for instance. That is, the powers of $X+Y$ can be expanded via the binomial expansion for complex numbers. In particular,

$$(X+Y)^n = \sum_{k=0}^{n} \binom{n}{k} X^{n-k} Y^k.$$

The exponential expansion is thus

$$e^{X+Y} = \sum_{m=0}^{\infty} \frac{X^m}{m!} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{k=0}^{m} \binom{m}{k} X^{m-k} Y^k.$$

Now consider the product $e^X e^Y$. Via the expansion, we have

$$e^X e^Y = \left(\sum_{p=0}^{\infty} \frac{X^p}{p!} \right) \left(\sum_{q=0}^{\infty} \frac{Y^q}{q!} \right).$$

For a fixed m, we can collected the terms of $e^X e^Y$ which are of power m by iteration; these terms are those that look like $X^m Y^0, X^{m-1} Y^1, \ldots X^0 Y^m$. For a given p and q, the coefficient in front of the $X^p Y^q$ term is explicitly given as $\frac{1}{p!q!}$. Hence, the terms of power m can be written as

$$\frac{1}{m!0!}X^mY^0 + \frac{1}{(m-1)!1!}X^{m-1}Y^1 + \cdots + \frac{1}{0!m!}X^0Y^m = \sum_{k=0}^{m}\frac{1}{m!}\binom{m}{k}X^{m-k}Y^k = \frac{1}{m!}\sum_{k=0}^{m}\binom{m}{k}X^{m-k}Y^k$$

We can then collect the terms as

$$e^Xe^Y = \left(\sum_{p=0}^{\infty}\frac{X^p}{p!}\right)\left(\sum_{q=0}^{\infty}\frac{Y^q}{q!}\right) = \sum_{m=0}^{\infty}\frac{1}{m!}\sum_{k=0}^{m}\binom{m}{k}X^{m-k}Y^k = e^{X+Y},$$

and the proof is complete since the equivalence can be shown up to arbitrary powers of m. ■

We can now decompose X as

$$X = \begin{pmatrix}3 & 0\\ 0 & 3\end{pmatrix} + \begin{pmatrix}1 & 3\\ -1 & -1\end{pmatrix} = 3I + \begin{pmatrix}1 & 3\\ -1 & -1\end{pmatrix} = A + B.$$

Clearly A and B are commutative since all matrices commute with the identity. We can compute the individual products e^A and e^B as follows. First,

$$e^A = \sum_{m=0}^{\infty}\frac{(3I)^n}{n!} = I\sum_{m=0}^{\infty}\frac{3^m}{m!} = e^3I = \begin{pmatrix}e^3 & 0\\ 0 & e^3\end{pmatrix}$$

and since B is traceless,

$$\begin{aligned}e^B &= \cos\left(\sqrt{\det B}\right)I + \frac{\sin\left(\sqrt{\det B}\right)}{\sqrt{\det B}}B = \cos\left(\sqrt{2}\right)I + \frac{\sin\left(\sqrt{2}\right)}{\sqrt{2}}B\\ &= \begin{pmatrix}\cos\sqrt{2} & 0\\ 0 & \cos\sqrt{2}\end{pmatrix} + \frac{\sin\sqrt{2}}{\sqrt{2}}\begin{pmatrix}1 & 3\\ -1 & -1\end{pmatrix}\\ &= \begin{pmatrix}\cos\sqrt{2} + \frac{\sin\sqrt{2}}{\sqrt{2}} & \frac{3\sin\sqrt{2}}{\sqrt{2}}\\ -\frac{\sin\sqrt{2}}{\sqrt{2}} & \cos\sqrt{2} - \frac{\sin\sqrt{2}}{\sqrt{2}}\end{pmatrix}\end{aligned}$$

and hence

$$\begin{aligned}e^X = e^{A+B} = e^Ae^B &= \begin{pmatrix}e^3 & 0\\ 0 & e^3\end{pmatrix}\begin{pmatrix}\cos\sqrt{2} + \frac{\sin\sqrt{2}}{\sqrt{2}} & \frac{3\sin\sqrt{2}}{\sqrt{2}}\\ -\frac{\sin\sqrt{2}}{\sqrt{2}} & \cos\sqrt{2} - \frac{\sin\sqrt{2}}{\sqrt{2}}\end{pmatrix}\\ &= e^3\begin{pmatrix}\cos\sqrt{2} + \frac{\sin\sqrt{2}}{\sqrt{2}} & \frac{3\sin\sqrt{2}}{\sqrt{2}}\\ -\frac{\sin\sqrt{2}}{\sqrt{2}} & \cos\sqrt{2} - \frac{\sin\sqrt{2}}{\sqrt{2}}\end{pmatrix},\end{aligned}$$

as was desired. ■

8. A matrix A is said to be unipotent if $A-I$ is nilpotent. Note that $\log A$ is defined whenever A is unipotent, because the series expansion terminates.

(a) Show that if A is unipotent, then $\log A$ is nilpotent.

Proof. By definition, we can write $A - I = N$, where N is nilpotent. Take some arbitrary l such that $n \geq l$ implies that $N^l = 0$. We can write

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{N^m}{m} = \sum_{m=1}^{l-1} (-1)^{m+1} \frac{N^m}{m}.$$

Now consider the product $(\log A)^2$. We have

$$(\log A)^2 = \left(N - \frac{N^2}{2} + \cdots\right)\left(N - \frac{N^2}{2} + \cdots\right) = N^2 + \mathcal{O}(N^3).$$

By induction, it follows that $(\log A)^n = N^n + \mathcal{O}(N^{n+1})$ so that $(\log A)^l = N^l + \mathcal{O}(N^{l+1}) = 0 + 0 = 0$ and hence $\log A$ is nilpotent. ■

(b) Show that if X is nilpotent, then e^X is unipotent.

Proof. Taking $X^l = 0$, we can write

$$e^X - I = \left(\sum_{m=0}^{\infty} \frac{X^m}{m!}\right) - I = \left(\sum_{m=0}^{l-1} \frac{X^m}{m!}\right) - I = \sum_{m=1}^{l-1} \frac{X^m}{m!} = X + \mathcal{O}(X^2).$$

By the inductive argument in part (a), it follows that $(e^X - I)^l = X^l + \mathcal{O}(X^{l+1}) = 0 + 0 = 0$ so that e^X is unipotent. ■

(c) Show that if A is unipotent, then $\exp(\log A) = A$ and that if X is nilpotent, then $\log(\exp X) = X$.

12. Show that two isomorphic matrix Lie groups have isomorphic Lie algebras.

Proof. Suppose $G \cong H$ and let $\Phi : G \to H$ be a Lie group isomorphism. Φ is by definition a Lie group homomorphism, which is additionally bijective. Let $\mathfrak{g}$ and $\mathfrak{h}$ denote the Lie algebras of G and H, respectively. By Theorem 2.21, there exists a linear map $\phi : \mathfrak{g} \to \mathfrak{h}$ such that

$$\phi(e^X) = e^{\phi(x)}$$

for all $X \in \mathfrak{g}$. We claim that ϕ is a Lie group isomorphism. We may assume that the bilinear map $[X, Y]$ on $\mathfrak{g}$ is the commutator since G is a *matrix* Lie group. Take some $X, Y \in \mathfrak{g}$. ■

13. *The Lie algebra* $\mathsf{so}(3;1)$. Write out explicitly the general form of a 4×4 real matrix in $\mathsf{so}(3;1)$.

Proof. For an arbitrary $X \in M_4(\mathbb{R})$, we have

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{pmatrix}.$$

The condition stated in the text is that $X \in \mathsf{so}(3;1)$ if and only if $gX^{\mathrm{T}}g = -X$ where

$$g = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix}.$$

Explicitly,

$$\begin{aligned} gX^{\mathrm{T}}g &= \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \begin{pmatrix} x_{11} & x_{21} & x_{31} & x_{41} \\ x_{12} & x_{22} & x_{32} & x_{42} \\ x_{13} & x_{23} & x_{33} & x_{43} \\ x_{14} & x_{24} & x_{34} & x_{44} \end{pmatrix} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \begin{pmatrix} x_{11} & x_{21} & x_{31} & -x_{41} \\ x_{12} & x_{22} & x_{32} & -x_{42} \\ x_{13} & x_{23} & x_{33} & -x_{43} \\ x_{14} & x_{24} & x_{34} & -x_{44} \end{pmatrix} \\ &= \begin{pmatrix} x_{11} & x_{21} & x_{31} & -x_{41} \\ x_{12} & x_{22} & x_{32} & -x_{42} \\ x_{13} & x_{23} & x_{33} & -x_{43} \\ -x_{14} & -x_{24} & -x_{34} & x_{44} \end{pmatrix} \\ &= \begin{pmatrix} -x_{11} & -x_{12} & -x_{13} & -x_{14} \\ -x_{21} & -x_{22} & -x_{23} & -x_{24} \\ -x_{31} & -x_{32} & -x_{33} & -x_{34} \\ -x_{41} & -x_{42} & -x_{43} & -x_{44} \end{pmatrix} = -X. \end{aligned}$$

For a real number a, we have $a = -a$ if and only if $a = 0$. Hence, there are 6 degrees of freedom for X and we can write

$$X = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix},$$

where $a, b, c, d, e, f \in \mathbb{R}$ are arbitrary. Notably, this resembles the dual tensor formulation of electrodynamics. ■

16. *The Lie algebras* $\mathsf{su}(2)$ *and* $\mathsf{so}(3)$. Show that the real Lie algebras $\mathsf{su}(2)$ and $\mathsf{so}(3)$ are isomorphic.

Proof. It was shown in the text that $X \in \mathsf{su}(2)$ satisfies $\text{trace}(X) = 0$ and $X^\dagger = -X$ for $X \in M_2(\mathbb{C})$. Hence, we can characterize $\mathsf{su}(2)$ as

$$\mathsf{su}(2) = \left\{ \begin{pmatrix} x_1 i & x_2 + x_3 i \\ -x_2 + x_3 i & -x_1 i \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\}.$$

It was also show that $X \in \mathsf{so}(3)$ satisfies $X^T = -X$ with $X \in M_3(\mathbb{R})$. Hence,

$$\mathsf{so}(3) = \left\{ \begin{pmatrix} 0 & x_1 & x_2 \\ -x_1 & 0 & x_3 \\ -x_2 & -x_3 & 0 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\}.$$

Consider the map $\phi : \mathsf{su}(2) \to \mathsf{so}(3)$ given by

$$\phi \begin{pmatrix} x_1 i & x_2 + x_3 i \\ -x_2 + x_3 i & -x_1 i \end{pmatrix} = -2 \begin{pmatrix} 0 & x_1 & x_2 \\ -x_1 & 0 & x_3 \\ -x_2 & -x_3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2x_1 & -2x_2 \\ 2x_1 & 0 & -2x_3 \\ 2x_2 & 2x_3 & 0 \end{pmatrix}.$$

We will first show that ϕ is a Lie group homomorphism. Take $X, Y \in \mathsf{su}(2)$. Then

$$\begin{aligned}
\phi([X,Y]) &= \phi(XY - YX) \\
&= \phi\left[\begin{pmatrix} x_1 i & x_2 + x_3 i \\ -x_2 + x_3 i & -x_1 i \end{pmatrix} \begin{pmatrix} y_1 i & y_2 + y_3 i \\ -y_2 + y_3 i & -y_1 i \end{pmatrix} \right. \\
&\quad \left. - \begin{pmatrix} y_1 i & y_2 + y_3 i \\ -y_2 + y_3 i & -y_1 i \end{pmatrix} \begin{pmatrix} x_1 i & x_2 + x_3 i \\ -x_2 + x_3 i & -x_1 i \end{pmatrix} \right] \\
&= \phi\left[\begin{pmatrix} -x_1 y_1 + (x_2 + x_3 i)(-y_2 + y_3 i) & x_1 i(y_2 + y_3 i) - (x_2 + x_3 i) y_1 i \\ (-x_2 + x_3 i) y_1 i - x_1 i(-y_2 + y_3) i & (-x_2 + x_3 i)(y_2 + y_3 i) - x_1 y_1 \end{pmatrix} \right. \\
&\quad \left. - \begin{pmatrix} -x_1 y_1 + (-x_2 + x_3 i)(y_2 + y_3 i) & (x_2 + x_3 i) y_1 i - x_1 i(y_2 + y_3 i) \\ x_1 i(-y_2 + y_3 i) - (-x_2 + x_3 i) y_1 i & (x_2 + x_3 i)(-y_2 + y_3 i) - x_1 y_1 \end{pmatrix} \right] \\
&= \phi\left[2 \begin{pmatrix} (x_2 y_3 - x_3 y_2) i & (x_3 y_1 - x_1 y_3) + (x_1 y_2 - x_2 y_1) i \\ (x_1 y_3 - x_3 y_1) + (x_1 y_2 - x_2 y_1) i & (x_3 y_2 - x_2 y_3) i \end{pmatrix} \right] \\
&= 4 \begin{pmatrix} 0 & x_3 y_2 - x_2 y_3 & x_1 y_3 - x_3 y_1 \\ x_2 y_3 - x_3 y_2 & 0 & x_2 y_1 - x_1 y_2 \\ x_3 y_1 - x_1 y_3 & x_1 y_2 - x_2 y_1 & 0 \end{pmatrix}.
\end{aligned}$$

On the other hand,

$$
\begin{aligned}
[\phi(X),\phi(Y)] &= \phi(X)\phi(Y)-\phi(Y)\phi(X)\\
&= \begin{pmatrix} 0 & -2x_1 & -2x_2 \\ 2x_1 & 0 & -2x_3 \\ 2x_2 & 2x_3 & 0 \end{pmatrix}\begin{pmatrix} 0 & -2y_1 & -2y_2 \\ 2y_1 & 0 & -2y_3 \\ 2y_2 & 2y_3 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -2y_1 & -2y_2 \\ 2y_1 & 0 & -2y_3 \\ 2y_2 & 2y_3 & 0 \end{pmatrix}\begin{pmatrix} 0 & -2x_1 & -2x_2 \\ 2x_1 & 0 & -2x_3 \\ 2x_2 & 2x_3 & 0 \end{pmatrix}\\
&= \begin{pmatrix} -4x_1y_1-4x_2y_2 & -4x_2y_3 & 4x_1y_3 \\ -4x_3y_2 & -4x_1y_1-4x_3y_3 & -4x_1y_2 \\ 4x_3y_1 & -4x_2y_1 & -4x_2y_2-4x_3y_3 \end{pmatrix}\\
&\quad - \begin{pmatrix} -4x_1y_1-4x_2y_2 & -4x_3y_2 & 4x_3y_1 \\ -4x_2y_3 & -4x_1y_1-4x_3y_3 & -4x_2y_1 \\ 4x_1y_3 & -4x_1y_2 & -4x_2y_2-4x_3y_3 \end{pmatrix}\\
&= 4\begin{pmatrix} 0 & x_3y_2-x_2y_3 & x_1y_3-x_3y_1 \\ x_2y_3-x_3y_2 & 0 & x_2y_1-x_1y_2 \\ x_3y_1-x_1y_3 & x_1y_2-x_2y_1 & 0 \end{pmatrix} = \phi([X,Y]),
\end{aligned}
$$

as desired. The bijective nature of ϕ can be seen by the bijective mapping of $(x_1,x_2,x_3)\to(x_1,x_2,x_3)$ which is readily observed as both onto and 1-1. Hence, ϕ is a Lie group isomorphism and $\mathsf{su}(2)\cong\mathsf{so}(3)$. ■

20. If $\mathfrak{g}$ is a Lie algebra, then a subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ is called an ideal if $[X,H]\in\mathfrak{h}$ for all $X\in\mathfrak{g}$ and $H\in\mathfrak{h}$. If $\phi:\mathfrak{g}_1\to\mathfrak{g}_2$ is a Lie group homomorphism, show that $\ker\phi$ is an ideal in $\mathfrak{g}_1$.

Proof. First we show that $\ker\phi$ is a subalgebra of $\mathfrak{g}_1$. By definition, $\ker\phi=\{\phi(X)=0 : X\in\mathfrak{g}_1\}$, where 0 is understood as the 0 vector in $\mathfrak{g}_2$. Take some $X,Y\in\ker\phi$. Then

$$\phi([X,Y])=[\phi(X),\phi(Y)]=[0,0]=0,$$

since $[X,X]=0$ for $X\in\mathfrak{g}_2$. Hence, $[X,Y]\in\ker\phi$ so that $\ker\phi$ is a subalgebra of $\mathfrak{g}_1$. Now take some $X\in\mathfrak{g}_1$ and $Y\in\ker\phi$. We want to show that $[X,Y]\in\ker\phi$. We have

$$\phi([X,Y])=[\phi(X),\phi(Y)]=[\phi(X),0].$$

We will show that this is 0. By the Jacobi identity, for $A\in\mathfrak{g}_2$, and assuming without loss of generality that $[A,0]=B$, with $B\in\mathfrak{g}_2$,

$$
\begin{aligned}
0 &= [A,[0,0]]+[0,[0,A]]+[0,[A,0]]\\
&= [A,0]+[0,-B]+[0,B]\\
&= B-[0,B]+[0,B]=B,
\end{aligned}
$$

where we have used $[X,Y]=-[Y,X]$ for all $X,Y\in\mathfrak{g}_2$ and the bilinearity of $[\cdot,\cdot]$. Hence, since $\phi(X)\in\mathfrak{g}_2$, we have $\phi([X,Y])=0$ and so $[X,Y]\in\ker\phi$. Thus, $\ker\phi$ is an ideal of $\mathfrak{g}_1$. ■

24. If $\mathfrak{g}$ is a Lie algebra, the center of $\mathfrak{g}$ is the set of all $Z \in \mathfrak{g}$ such that $[X, Z] = 0$ for all $X \in \mathfrak{g}$. Show that the center of $\mathfrak{g}$ is an ideal.

Proof. We first show that $Z(\mathfrak{g})$ is a subalgebra of $\mathfrak{g}$. Take $X, Y \in Z(\mathfrak{g})$. Then by definition, $[X, Y] = 0$. Hence, for arbitrary $Z \in \mathfrak{g}$, we have $[Z, [X, Y]] = [Z, 0] = 0$ by Exercise 20, so that $[X, Y] \in Z(\mathfrak{g})$ as desired.

We now show that $Z(\mathfrak{g})$ is an ideal of $\mathfrak{g}$. Take $X \in \mathfrak{g}$ and $Y \in Z(\mathfrak{g})$. Then we have $[X, Y] = 0$ again by definition of $Z(\mathfrak{g})$. Hence, for arbitrary $Z \in \mathfrak{g}$, we have $[Z, [X, Y]] = [Z, 0] = 0$ once again by the methods of Exercise 20. Thus, $[X, Y] \in Z(\mathfrak{g})$ as desired and $Z(\mathfrak{g})$ is an ideal of $\mathfrak{g}$. ■